Lecture 08: Keys, Functional Dependencies, and Normalization

ITCS216 Fundamentals of Database Systems
Semester 1/2026

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This lecture is adopted from ITCS241 content in 1/2022 and collarborate with Asst. Prof. Dr. Preecha Tangworakitthaworn

Learning Objectives

After completing this lesson, you will be able to:

- Understand keys and functional dependencies.
- Understand the normalization and perform normalization processes.

Relational Database Keys

- Consist of one or more attributes of a table that determine other attributes in the different table.
- Used to:
  - Ensure that each row in a table is **uniquely identifiable,**
  - Establish relationships among tables and to ensure the integrity of the data.

Two main keys:

- **Primary key (PK):** Attribute or combination of attributes that uniquely identifies any given row.
- **Foreign Key (FK):** Attribute that identifies another attribute stored in another table.

Relational Database Keys

RELATIONAL DATABASE KEYS	
KEY TYPE	**DEFINITION**
Superkey	An attribute or combination of attributes that uniquely identifies each row in a table
Candidate key	A minimal (irreducible) superkey; a superkey that does not contain a subset of attributes that is itself a superkey
Primary key	A candidate key selected to uniquely identify all other attribute values in any given row; cannot contain null entries
Foreign key	An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null
Secondary key	An attribute or combination of attributes used strictly for data retrieval purposes

Superkey

- An attribute or combination of attributes that uniquely identifies each row in a table.
- Superkey is a univ
- .

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

What else can be superkey?

- In the table above, some examples of superkey are:
  - SSN
    - SSN is unique for every row of data
  - (SSN, FNAME)
    - FNAME of employee can be the same, but their SSN is unique.
      So the combination of these two attributes can also be a key.
  - (FNAME, LNAME)
    - Either FNAME and LNAME of employee can be the same.
      But the combination of these two attributes is unique. So this can also be a key.
  - Etc.

Candidate Key

- A minimal (irreducible) superkey; a superkey that does not contain a subset of attributes that is itself a superkey.
- Candidate key can act as a primary key for a table. There can be **more than one** candidate key.
- Candidate key is a subset of superkey.

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

- From the table above:
  - SSN and (FNAME, LNAME) are superkeys and also candidate keys.
  - (SSN, FNAME) is a superkey but is NOT a candidate key.

Primary Key (PK)

- A candidate key which is selected to uniquely identify all other attribute values in any given row.
- Cannot contain null entries.
- One table has **only one** primary key.

Null: Absence of any data value that could represent:
- An unknown attribute value
- A known, but missing, attribute value
- A inapplicable condition

SSN (PK)	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

- Suppose there are 2 possible candidate keys: SSN and (FNAME, LNAME)
  - We can select SSN from the candidate keys, and make it as a primary key.
- **Surrogate Key** is special type of PK that contains unique values automatically generated by the database system.
  - This key usually has no meaning, except uniquely identifying a record.

Secondary or Alternative Keys

- The candidate key which are **not** selected as primary key.
- An attribute or combination of attributes used strictly for data retrieval purposes.

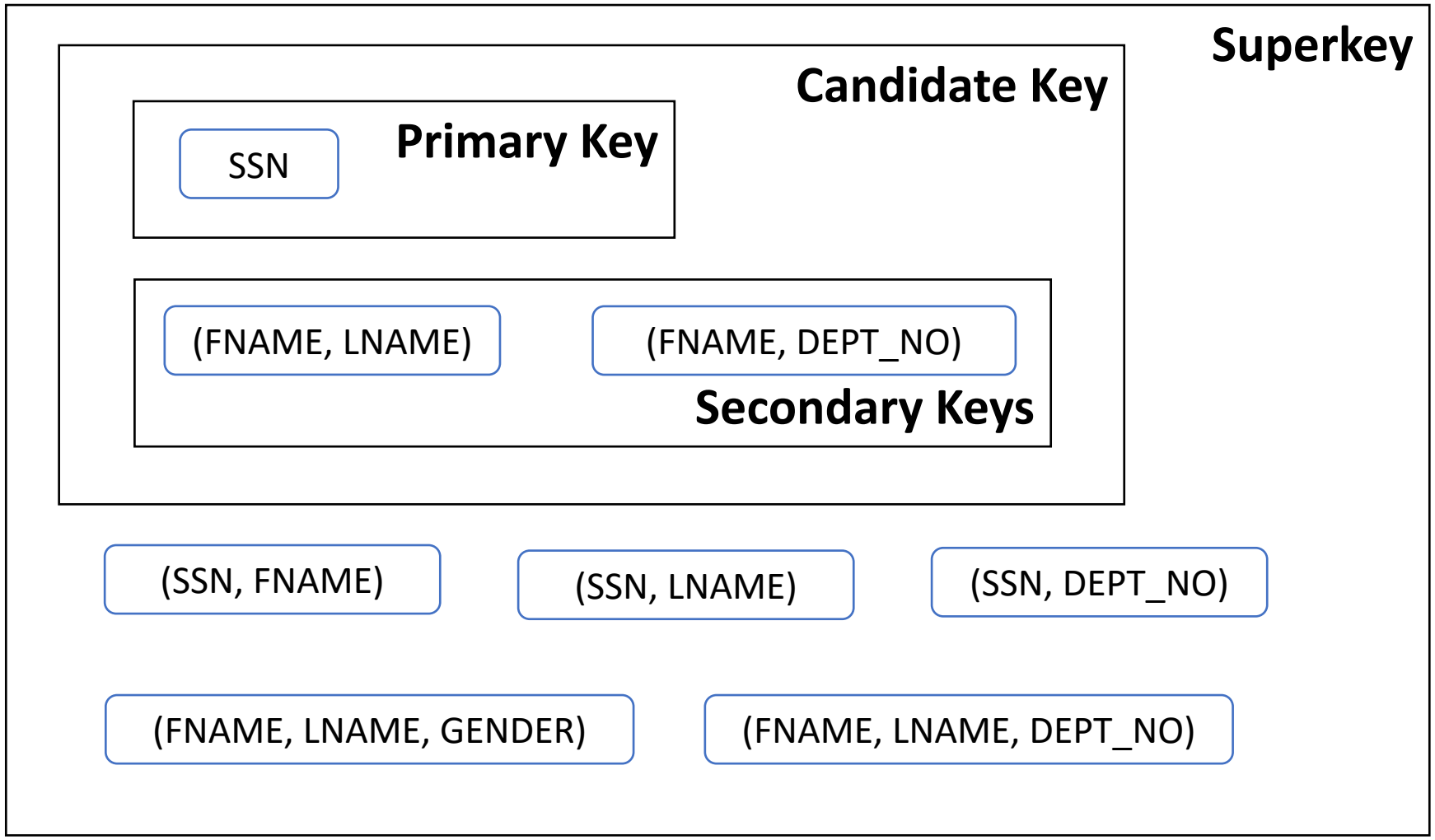
SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

- Suppose the candidate keys are SSN and (FNAME, LNAME):
  - If we select SSN as a primary key,

    then (FNAME, LNAME) is the secondary key.

Example Keys

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1



Foreign Key

- An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null.
- Suppose this is the data we want to store.

SSN	FNAME	LNAME	GENDER	DEPT_NO	DEPT_NAME	LOCATION
999887777	Alicia	Wallace	F	4	Finance	Los Angeles
987654321	Alicia	Zelaya	F	5	IT	San Francisco
666884444	Ramesh	Narayan	M	5	IT	San Francisco
453453453	Bob	Wallace	M	1	Admin	Los Angeles

We can create a better design using relational data model as shown below

Employee

FK

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

PK

Department

DEPT_NO	NAME	LOCATION
1	Admin	Los Angeles
2	Human Resource	Los Angeles
3	Marketing	Los Angeles
4	Finance	Los Angeles
5	IT	San Francisco

Integrity Rules

- **Entity integrity:** Condition in which each row in the table has its own unique identity.
  - All of the values in the **primary key** must be unique.
  - No key attribute in the **primary key** can contain a null value.

- **Referential integrity:** Every reference to an entity instance by another entity instance is valid.
  - Every non-null **foreign key** value must reference an existing primary key value.
  - A key attribute in the **foreign key** can contain a null value.

Integrity Rules: Entity Integrity

ENTITY INTEGRITY	DESCRIPTION
Requirement	All primary key entries are unique, and no part of a primary key may be null.
Purpose	Each row will have a unique identity, and foreign key values can properly reference primary key values.
Example	No invoice can have a duplicate number, nor can it be null; in short, all invoices are uniquely identified by their invoice number.

Employee

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

Cannot insert this record because the SSN value is duplicate

453453453	Bobby	Wallace	M	1

Cannot insert this record because the SSN value is NULL or missing

	Alex	Irvine	M	2

Can we insert these new records?

Integrity Rules: Referential Integrity

REFERENTIAL INTEGRITY	DESCRIPTION
Requirement	A foreign key may have either a null entry, as long as it is not a part of its table's primary key, or an entry that matches the primary key value in a table to which it is related; (every non-null foreign key value *must* reference an *existing* primary key value).
Purpose	It is possible for an attribute *not* to have a corresponding value, but it will be impossible to have an invalid entry; the enforcement of the referential integrity rule makes it impossible to delete a row in one table whose primary key has mandatory matching foreign key values in another table.
Example	A customer might not yet have an assigned sales representative (number), but it will be impossible to have an invalid sales representative (number).

Employee

FK

SSN	FNAME	LNAME	GENDER	DEPT_NO
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1
123456789	Alex	Irvine	M	2
123455555	Cora	Jay	F	
987655555	Elsa	Arendelle	F	10

Null value is allowed

Invalid key value

PK

Department

DEPT_NO	NAME	LOCATION
1	Admin	Los Angeles
2	Human Resource	Los Angeles
3	Marketing	Los Angeles
4	Finance	Los Angeles
5	IT	San Francisco

Can we insert these new records?

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Functional Dependency



Functional Dependency

- Constraint on the possible rows in a table.
- Value neutral like FKs and PKs.
- Require understanding of business rules.

Functional Dependencies (FDs)

- $X \rightarrow Y$
- X (functionally) determines Y
- X: left-hand-side (LHS) or determinant set
- Y: right-hand-side (RHS) or dependent set
- For each X value, there is at most one Y value
- Similar to identifying potential candidate key

We would read this as, “Y is dependent on X”, or “X determines the value of Y uniquely”

For example, **StudentID → StudentName**

StudentID is the determinant attribute and **StudentName** is the dependent attribute

FD Examples (multiple attributes)

- $X_1 \rightarrow Y_1$
  - StudentID → StudentName
- $X \rightarrow (Y_1, Y_2, Y_3)$
  - StudentID → (StudentName, email, CGPA)
- $(X_1, X_2) \rightarrow Y_1$
  - (email, phoneNumber) → StudentID
- $(X_1, X_2) \rightarrow (Y_1, Y_2, Y_3)$
  - (StudentID, StudentName) → (email, phoneNumber, CGPA)

FDs More Example from a Relation

Student (StdID, sName, address, HScode, HSname, HScity, GPA, honor)

- StdID → sName
- StdID → address
- HSCode → HSname, HSCity
- HSname, HSCity → HScode
- StdID → GPA
- GPA → honor
- StdID → honor

Note that: The last FD can be derived from *Transitive Rule*
If A→ B and B → C then A → C

And More ….

Implementation of Functional Dependency



How to use functional dependency in SQL.

if we execute the following query, we should always get only one student_name for a given reg_no.

"SELECT Student_Name FROM Student WHERE Reg_No = 123;"

FD1: REG_NO → STUDENT_NAME

This functional dependency means that there is at most one student name is related to one register number. This is correct.

FD2: STUDENT_NAME → REG_NO

This functional dependency means that there is at most one register number is related to a student name. This may not be true. Because, there may be more than one student with the same name.

Implementation of Functional Dependency

Given a relation R (A, B, C, D) with the following instances:

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Check the functional dependencies of a relation for the following FDs:

(a) A → B
✓ (b) A → CD
✓ (c) AB → CD
✓ (d) C → D
✓ (e) B → A
✓ (f) BD → AC
(g) AD → BC
✓ (h) D → B
✓ (i) D → C
✓ (j) C → A

Implementation of Functional Dependency

(a) A → B

Given a relation R (A, B, C, D) with the following instances:

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Solution: A→B does not hold in relation R.

WHY? In the table R, we have 3 B values (1, 2 and 3) for a given A value (1).

Implementation of Functional Dependency

(b) A → CD

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Solution: A→CD holds in relation R.

WHY? In R, we have single C and D combination of values for every A value. You can observe the following from R.

A	CD
1	(2,3)
2	(5,6)
5	(7,8)

Implementation of Functional Dependency

(c) AB → CD

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Solution: AB→CD holds in relation R.

WHY? In R there is one to one relationship between AB and CD, i.e., for a given A and B combination of values, there is unique C and D combination of values.

AB	CD
(1,1)	(2,3)
(1,2)	(2,3)
(1,3)	(2,3)
(2,4)	(5,6)
(5,6)	(7,8)

Implementation of Functional Dependency

(d) C → D

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Solution: C→D holds in relation R.

WHY? In R there is one to one relationship between C and D values.

C	D
2	3
5	6
7	8

Implementation of Functional Dependency

(e) B → A

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

Solution: B→A holds in relation R.

WHY? In R there is one to one relationship between B and A values from B. that is, for a given B value, there is only one associated A value.

B	A
1	1
2	1
3	1
4	2
6	5

Implementation of Functional Dependency

Try yourself!!

A	B	C	D
1	1	2	3
1	2	2	3
1	3	2	3
2	4	5	6
5	6	7	8

(f) BD → AC

Your solution is holds

(g) AD → BC

Your solution is does not hold

(h) D → B

Your solution is does not hold

(i) D → C

Your solution is holds

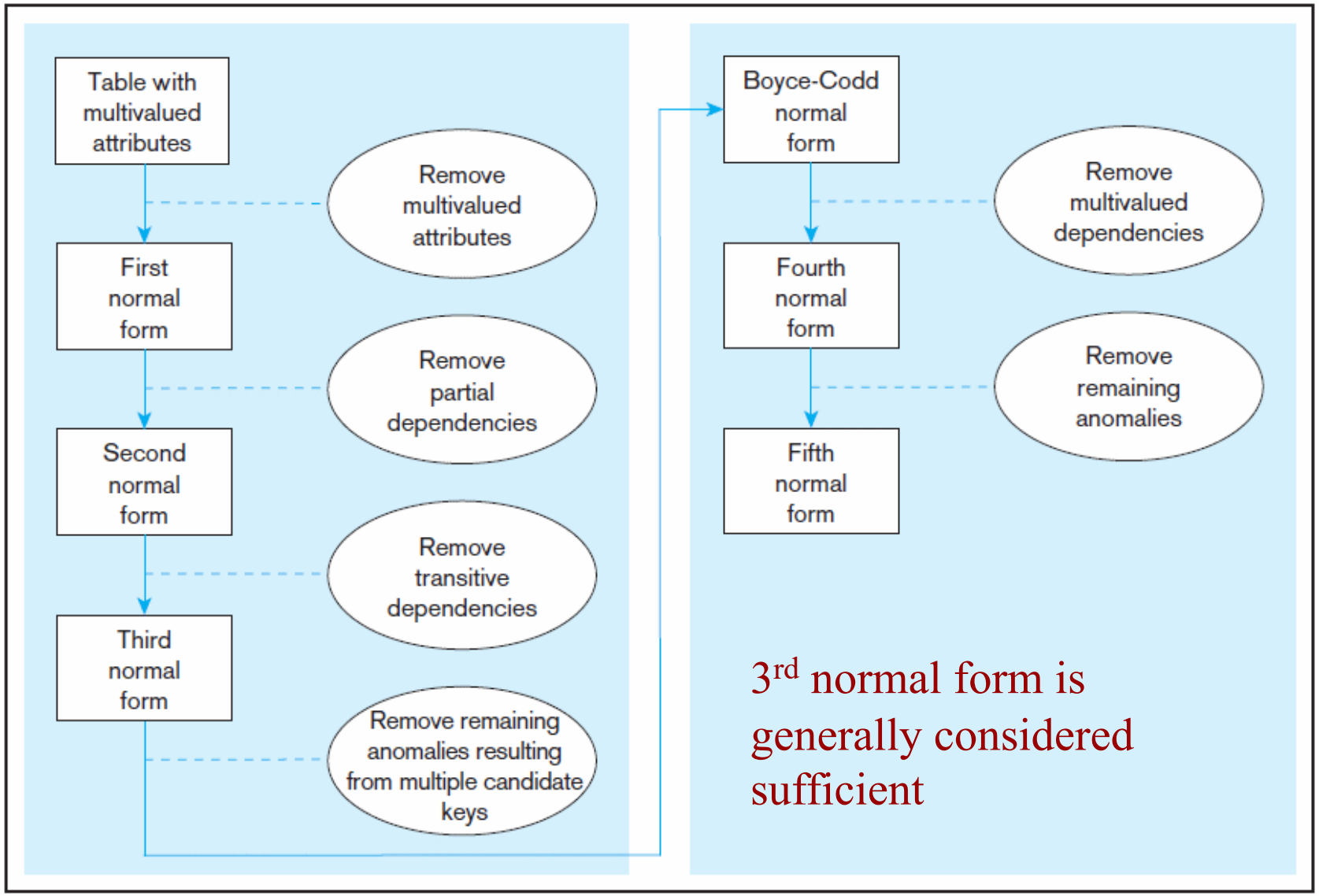
(j) C → A

Your solution is holds

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Normalization Process

Figure 4.22 Steps in normalization



Normalization Process

Objective is to ensure that each table conforms to the concept of well-formed and well-designed relations:

- Each table represents a single subject domain (or data domain),
- No data item will be unnecessarily stored in more than one table,
- All nonprime (or non-key) attributes in a table are dependent on the primary key,
- Each table is void of insertion, update, and deletion anomalies.

Normalization Process

- Ensures that all tables are in at least 3NF.
- Higher forms are not likely to be encountered in business environment.
- Works one relation at a time.
- Starts by:
  - Identifying the dependencies of a relation (table),
  - Progressively breaking the relation into new set of relations.

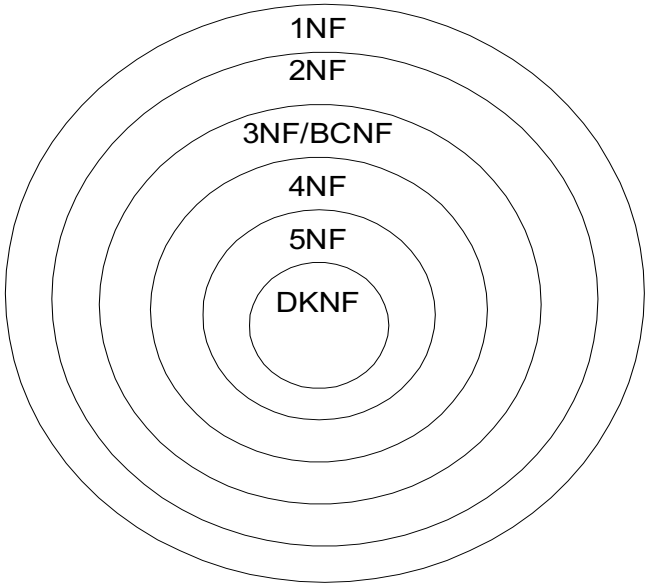
Normal Forms

TABLE 6.2

NORMAL FORMS

NORMAL FORM	CHARACTERISTIC	SECTION
First normal form (1NF)	Table format, no repeating groups, and PK identified	6.3.1
Second normal form (2NF)	1NF and no partial dependencies	6.3.2
Third normal form (3NF)	2NF and no transitive dependencies	6.3.3
Boyce-Codd normal form (BCNF)	Every determinant is a candidate key (special case of 3NF)	6.6.1
Fourth normal form (4NF)	3NF and no independent multivalued dependencies	6.6.2

Relationships of Normal Forms



Conversion to First Normal Form

- **Repeating group**: Group of multiple entries of same type can exist for any single key attribute occurrence
  - Existence proves the presence of data redundancies
- Enable reducing data redundancies
- 3 main steps of 1NF:
  - Eliminate the repeating groups
  - Identify the primary key
  - Identify all dependencies

Gr	Business	ID
1	Banking	5888037
		5888068
		5888079
		5888119
2	Airlines	5888139
		5888203
		5888204
		5888251

Conversion to First Normal Form

Student Relation

Gr	Business	ID
1	Banking	5888037, 5888068, 5888079, 5888119
2	Airlines	5888139, 5888203, 5888204, 5888251



Results after Normalization (Eliminating Repeating Group)

Gr	Business	ID
1	Banking	5888037
1	Banking	5888068
1	Banking	5888079
1	Banking	5888119
2	Airlines	5888139
2	Airlines	5888203
2	Airlines	5888204
2	Airlines	5888251

Repeating group is Multi-valued construction in database table.

Conversion to First Normal Form

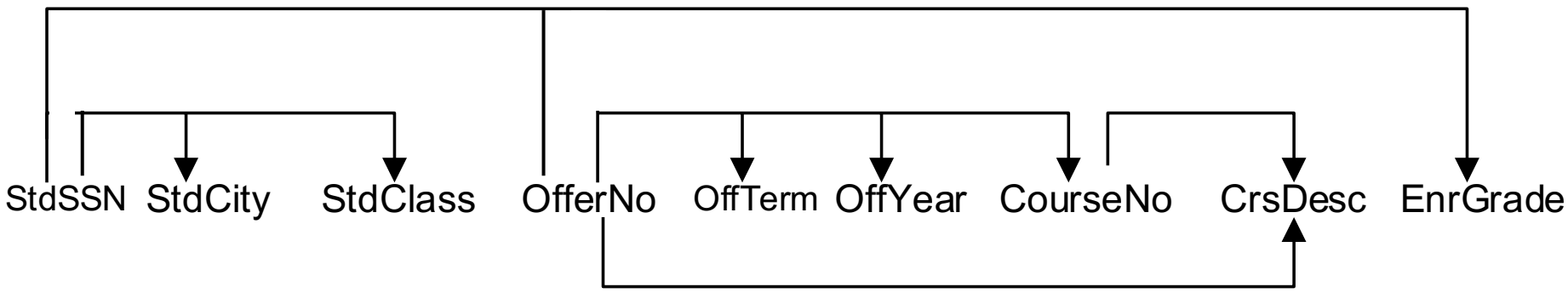
Dependency diagram: Depicts all dependencies found within given table structure:

- Helps to get an overview of all relationships among table's attributes,
- Makes it less likely that an important dependency will be overlooked.

Conversion to First Normal Form

- 1NF describes tabular format in which:
  - All key attributes are defined,
  - There are no repeating groups in the table,
  - All attributes are dependent on the primary key.
- All relational tables satisfy 1NF requirements.
- Some tables contain partial dependencies.
  - Subject to data redundancies and various anomalies.

Dependency Diagram



For example:

StdSSN → StdCity, StdClass

OfferNo → OffTerm, OffYear, CourseNo, CrsDesc

CourseNo → CrsDesc

StdSSN, OfferNo → EnrGrade

Table with multivalued attributes, not in 1st normal form

OrderID	Order Date	Customer ID	Customer Name	Customer Address	ProductID	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
					5	Writer's Desk	Cherry	325.00	2
					4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4–Dr Dresser	Oak	500.00	4
					4	Entertainment Center	Natural Maple	650.00	3

FIGURE 4-25 INVOICE data (Pine Valley Furniture Company)

Note: This is NOT a relation. WHY?

Table with no multivalued attributes and unique rows, in 1st normal form

OrderID	Order Date	Customer ID	Customer Name	Customer Address	ProductID	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	5	Writer's Desk	Cherry	325.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4–Dr Dresser	Oak	500.00	4
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	4	Entertainment Center	Natural Maple	650.00	3

FIGURE 4-26 INVOICE relation (1NF) (Pine Valley Furniture Company)

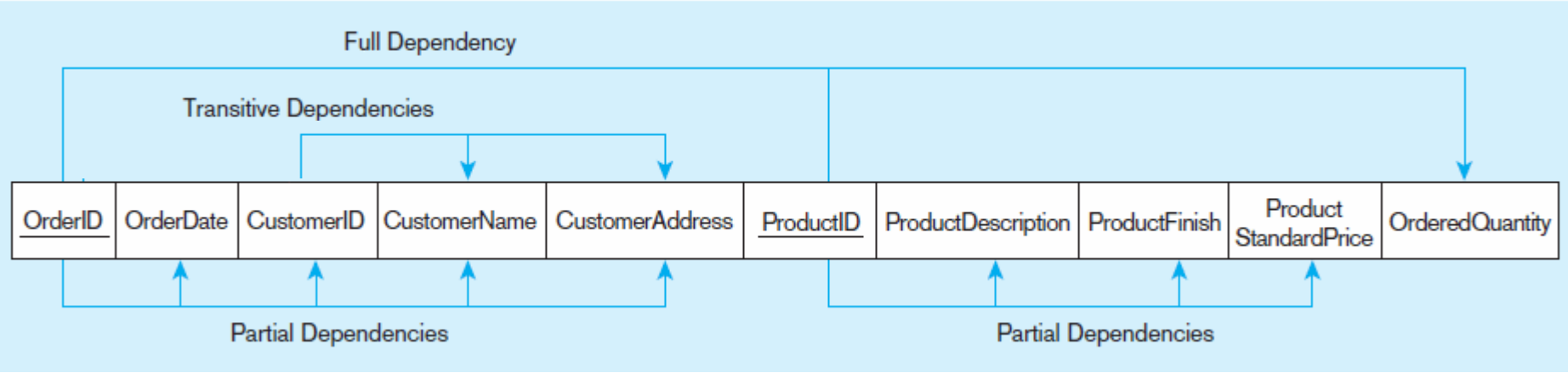
Note: This is a relation, but not a well-structured one.

Conversion to Second Normal Form

- Steps:
  - Make new tables to eliminate partial dependencies,
  - Reassign corresponding dependent attributes.
- Table is in 2NF when it:
  - is in 1NF,
  - no partial dependencies maintained in the relation

    So, removed partial dependencies.

Functional dependency diagram for INVOICE



OrderID ➔ OrderDate, CustomerID, CustomerName, CustomerAddress

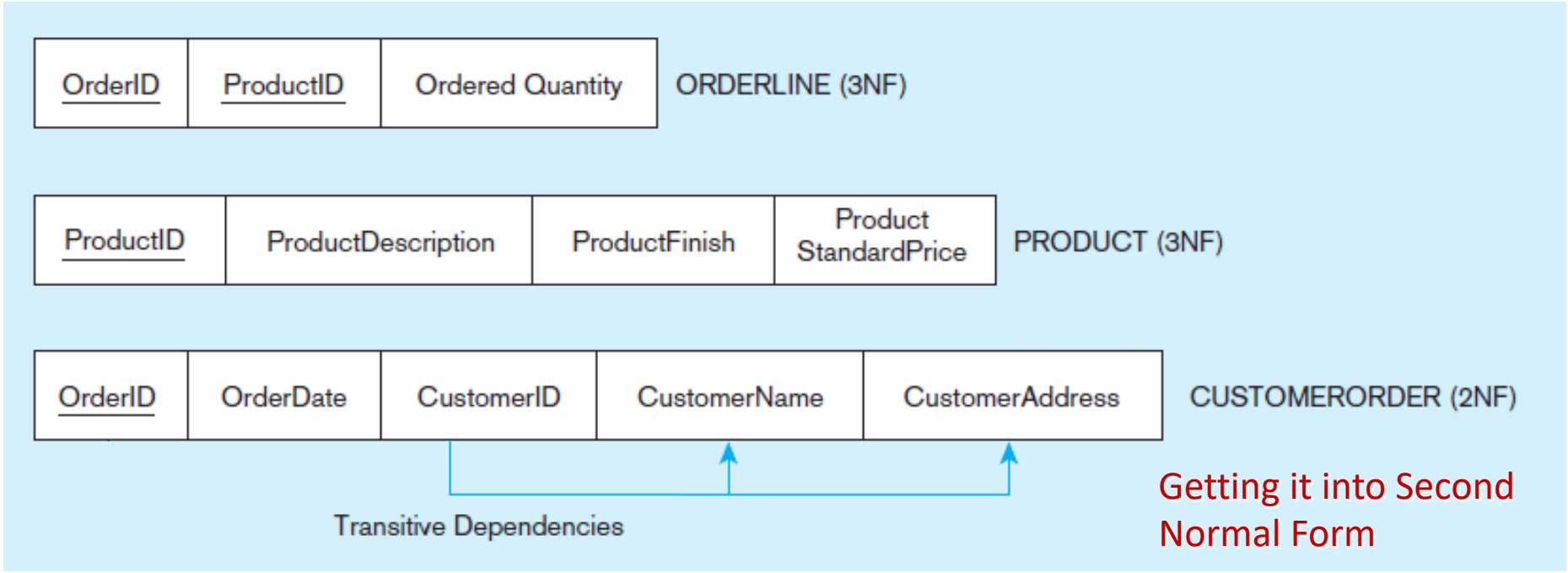
CustomerID ➔ CustomerName, CustomerAddress

ProductID ➔ ProductDescription, ProductFinish, ProductStandardPrice

OrderID, ProductID ➔ OrderQuantity

Therefore, NOT in 2nd Normal Form

Removing partial dependencies



Partial dependencies are removed, but there still cover transitive dependency!

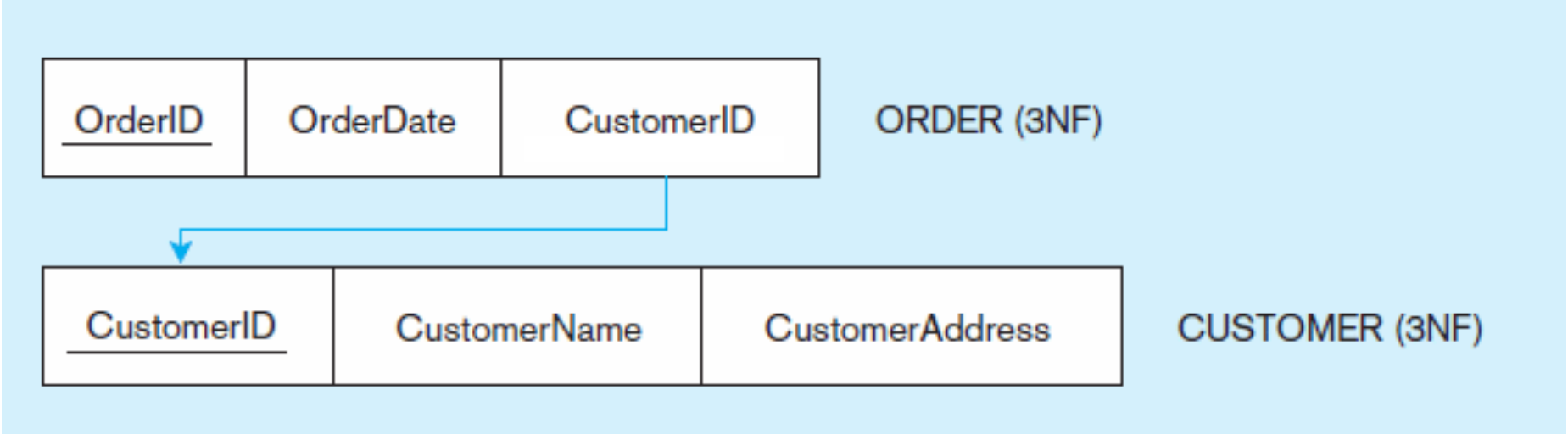
Note that, a partial dependency **can exist only** when a table's primary key is composed of several attributes, a table with single attribute primary key is automatically in 2NF once it is 1NF.

Conversion to Third Normal Form

- Steps:
  - Make new tables to eliminate transitive dependencies.
    - Write a copy of its determinant as a primary key for a new table (**Determinant**: Any attribute whose value determines other values within a row).
  - Reassign corresponding dependent attributes.
- Table is in 3NF when it:
  - is in 2NF,
  - no transitive dependencies maintained in the relation.

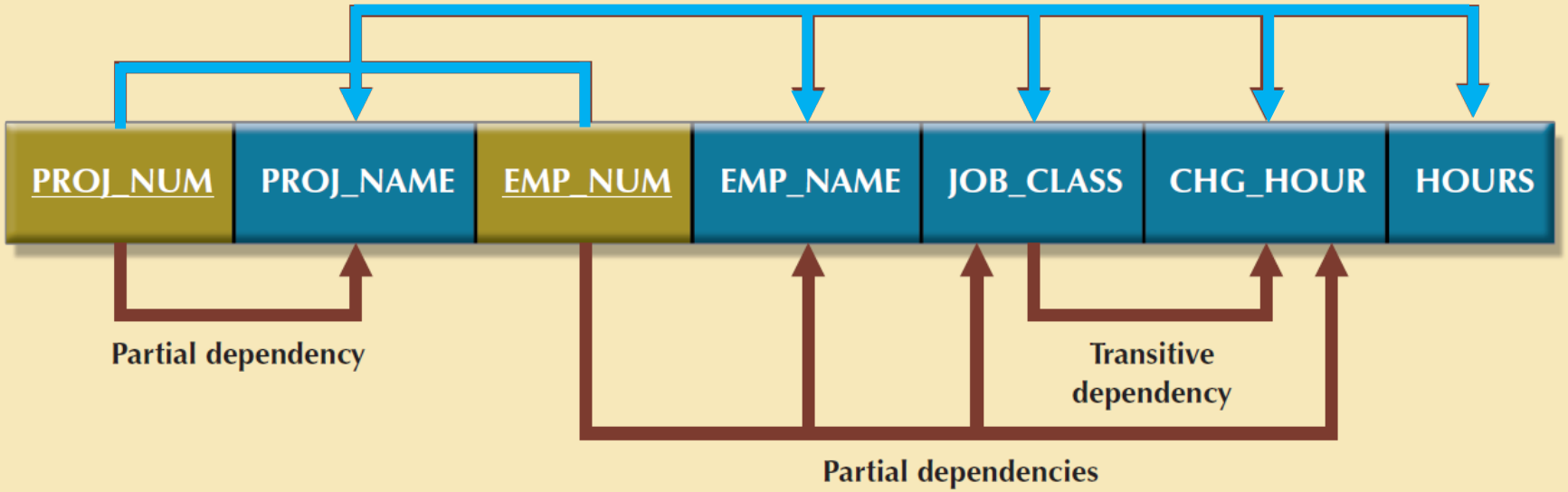
    So, remove transitive dependencies.

Removing transitive dependencies



Transitive dependencies are removed.

Another Example: Company Database
First Normal Form (1NF) Dependency Diagram



1NF (PROJ_NUM, EMP_NUM, PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOURS, HOURS)

PARTIAL DEPENDENCIES:
(PROJ_NUM → PROJ_NAME)
(EMP_NUM → EMP_NAME, JOB_CLASS, CHG_HOUR)

TRANSITIVE DEPENDENCY:
(JOB_CLASS → CHG_HOUR)

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Second Normal Form (2NF) Conversion Results

Table name: PROJECT

PROJECT (PROJ_NUM, PROJ_NAME)

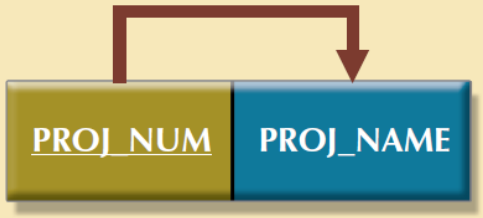


Table name: EMPLOYEE

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS, CHG_HOUR)

TRANSITIVE DEPENDENCY
(JOB_CLASS → CHG_HOUR)

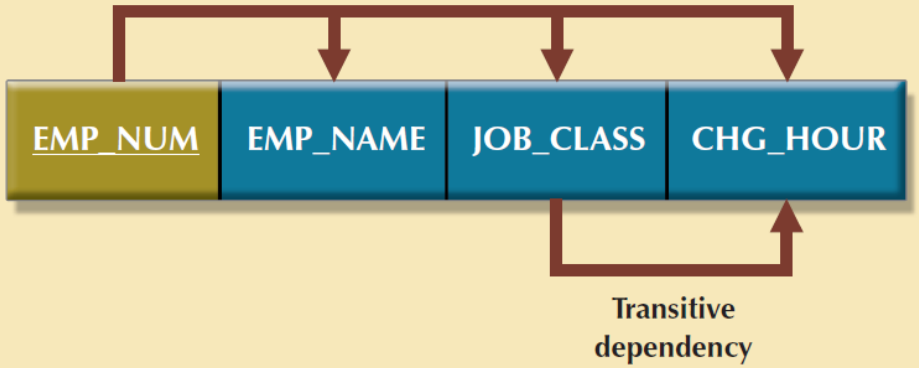
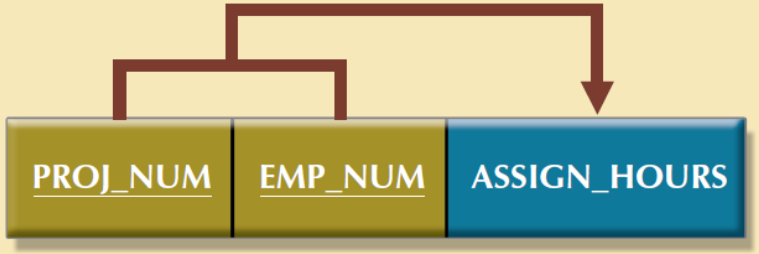


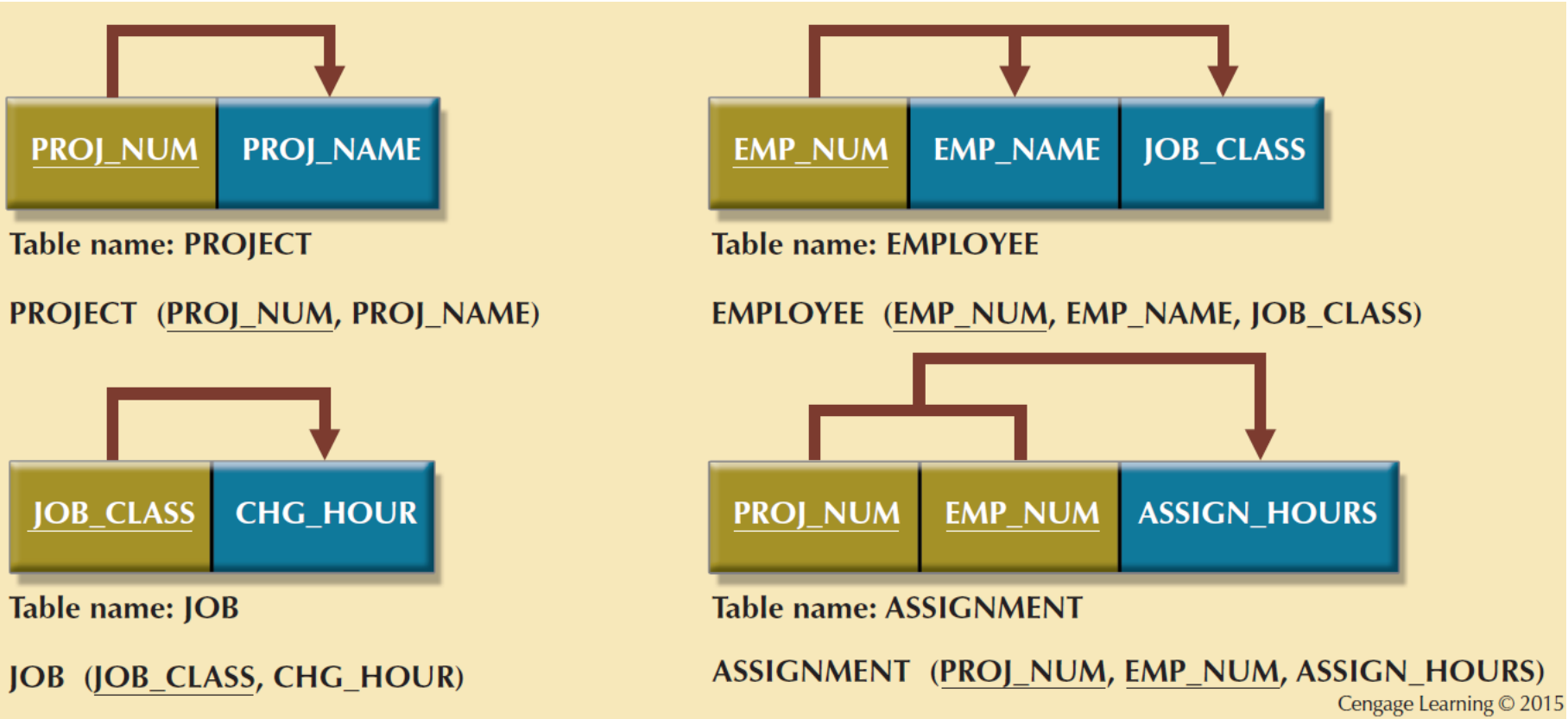
Table name: ASSIGNMENT

ASSIGNMENT (PROJ_NUM, EMP_NUM, ASSIGN_HOURS)



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Third Normal Form (3NF) Conversion Results



The Complete Database

Enhancing Design after 3NF

Table name: PROJECT

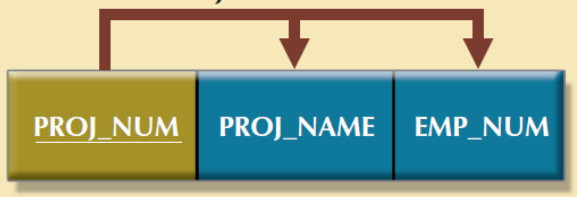


Table name: PROJECT

PROJ_NUM	PROJ_NAME	EMP_NUM
15	Evergreen	105
18	Amber Wave	104
22	Rolling Tide	113
25	Starflight	101

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Table name: JOB Database name: Ch06_ConstructCo

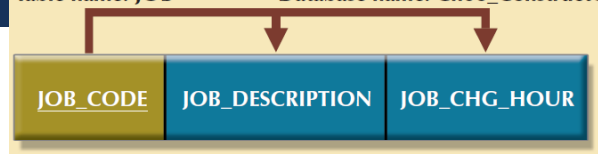


Table name: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
500	Programmer	35.75
501	Systems Analyst	96.75
502	Database Designer	105.00
503	Electrical Engineer	84.50
504	Mechanical Engineer	67.90
505	Civil Engineer	55.78
506	Clerical Support	26.87
507	DSS Analyst	45.95
508	Applications Designer	48.10
509	Bio Technician	34.55
510	General Support	18.36

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Table name: ASSIGNMENT

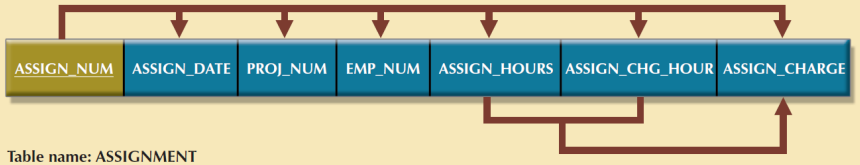


Table name: ASSIGNMENT

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-14	15	103	2.6	84.50	219.70
1002	04-Mar-14	18	118	1.4	18.36	25.70
1003	05-Mar-14	15	101	3.6	105.00	378.00
1004	05-Mar-14	22	113	2.5	48.10	120.25
1005	05-Mar-14	15	103	1.9	84.50	160.55
1006	05-Mar-14	25	115	4.2	96.75	406.35
1007	05-Mar-14	22	105	5.2	105.00	546.00
1008	05-Mar-14	25	101	1.7	105.00	178.50
1009	05-Mar-14	15	105	2.0	105.00	210.00
1010	06-Mar-14	15	102	3.8	96.75	367.65
1011	06-Mar-14	22	104	2.6	96.75	251.55
1012	06-Mar-14	15	101	2.3	105.00	241.50
1013	06-Mar-14	25	114	1.8	48.10	86.58
1014	06-Mar-14	22	111	4.0	26.87	107.48
1015	06-Mar-14	25	114	3.4	48.10	163.54
1016	06-Mar-14	18	112	1.2	45.95	55.14
1017	06-Mar-14	18	118	2.0	18.36	36.72
1018	06-Mar-14	18	104	2.6	96.75	251.55
1019	06-Mar-14	15	103	3.0	84.50	253.50
1020	07-Mar-14	22	105	2.7	105.00	283.50
1021	08-Mar-14	25	108	4.2	96.75	406.35
1022	07-Mar-14	25	114	5.8	48.10	278.98
1023	07-Mar-14	22	106	2.4	35.75	85.80

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Table name: EMPLOYEE Database name: Ch06_ConstructCo

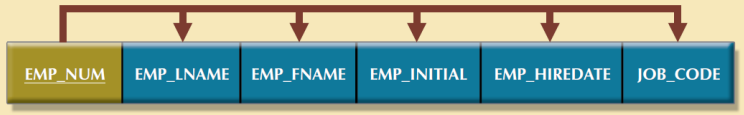


Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-00	502
102	Senior	David	H	12-Jul-89	501
103	Arbough	June	E	01-Dec-97	503
104	Ramoras	Anne	K	15-Nov-88	501
105	Johnson	Alice	K	01-Feb-94	502
106	Smithfield	William		22-Jun-05	500
107	Alonzo	Maria	D	10-Oct-94	500
108	Washington	Ralph	B	22-Aug-89	501
109	Smith	Larry	W	18-Jul-99	501
110	Olenko	Gerald	A	11-Dec-96	505
111	Wabash	Geoff	B	04-Apr-89	506
112	Smithson	Darlene	M	23-Oct-95	507
113	Joenbrood	Delbert	K	15-Nov-94	508
114	Jones	Annelise		20-Aug-91	508
115	Bawangi	Travis	B	25-Jan-90	501
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-06	510

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